High-Optical-Depth Cold Atom Ensemble for Narrowband Entangled Photon Generation

4/16/26

Presenting: Ian Jack

PI: Shengwang Du

Graduate students: Xuanying Lai, Christopher Li, Adi Bhatnagar

PURDUE UNIVERSITY®

AGENDA

01

Motivation

General purpose of the experiment in the context of quantum computing

02

Background

Necessary information from AMO, quantum computing, and physics

03

Methods

Techniques and resources used to realize the experiment

04

Results

Presentation and brief analysis of numerical results

05

Next Steps

What can be done next to further explore or apply the knowledge

PURDUE UNIVERSITY

Motivation

PURDUE UNIVERSITY®

Applications of Quantum Computing

- Complex modelling
  - Drug models
  - Physical chemistry models
  - Financial/Complex System Models
- Optimization Problems
  - Manufacturing flow
  - Traffic reduction
- Encryption
- Overall creation of a **quantum advantage** in these problems

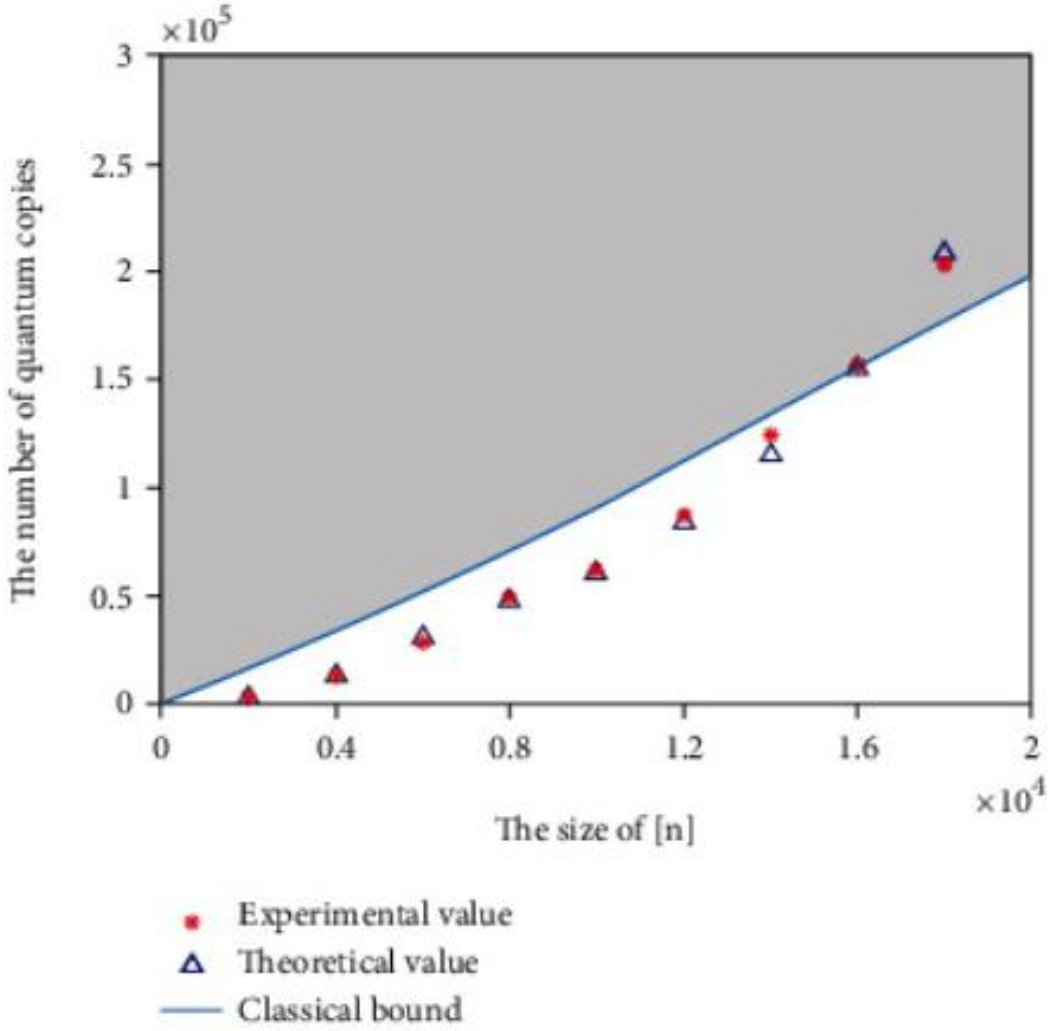

Figure 1: Quantum advantage visualized[7]

PURDUE UNIVERSITY

Benefits of MOTs in Distributed Quantum Computing

- <u>**Distributed quantum computing**</u> architecture is more cost-effective to scale
- This requires information to travel long distances without <u>**decohering**</u>
- Creating MOTs allows for coherent states to travel through fiber cables over potentially long distances

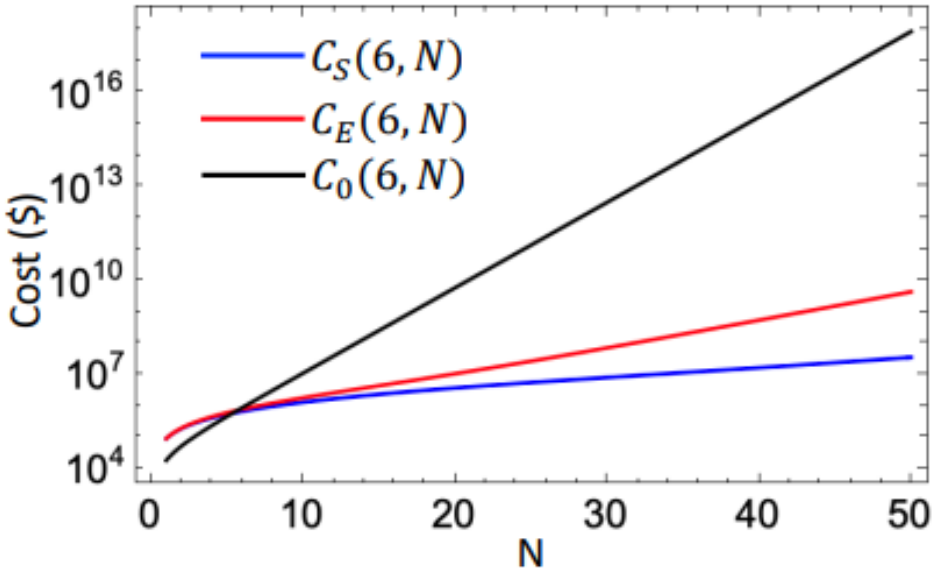

Figure 2: Distributed quantum computing cost per qubit comparison[3]

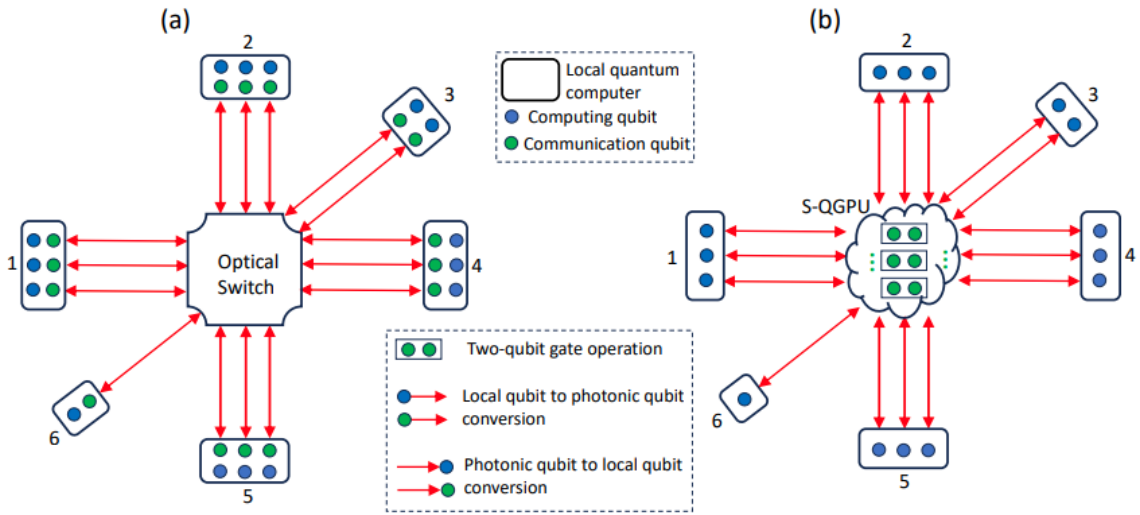

Figure 3: Distributed quantum computing architecture[3]

PURDUE UNIVERSITY

Background

PURDUE UNIVERSITY®

What is a quantum computer?

- <u>**Superposition**</u> for efficacy in some calculation
- Perform operations using <u>**quantum circuits**</u>
- Leveraged in many different platforms
  - Superconductors
  - Topological
  - <u>Neutral atom</u>
  - Photonic

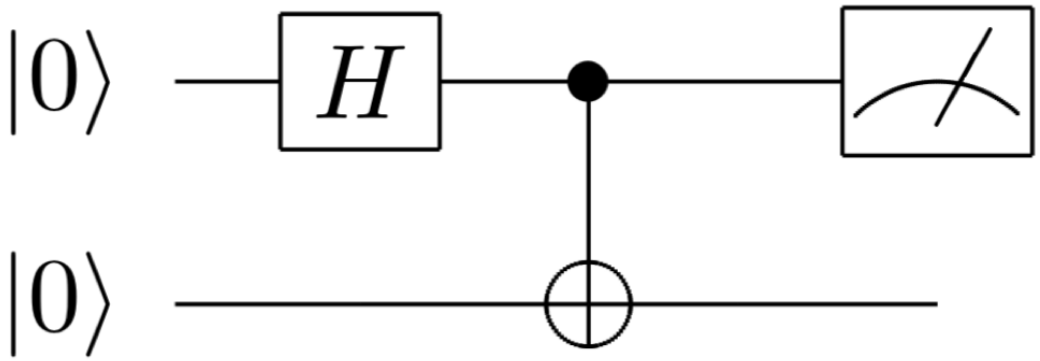

Figure 4: Quantum circuit example

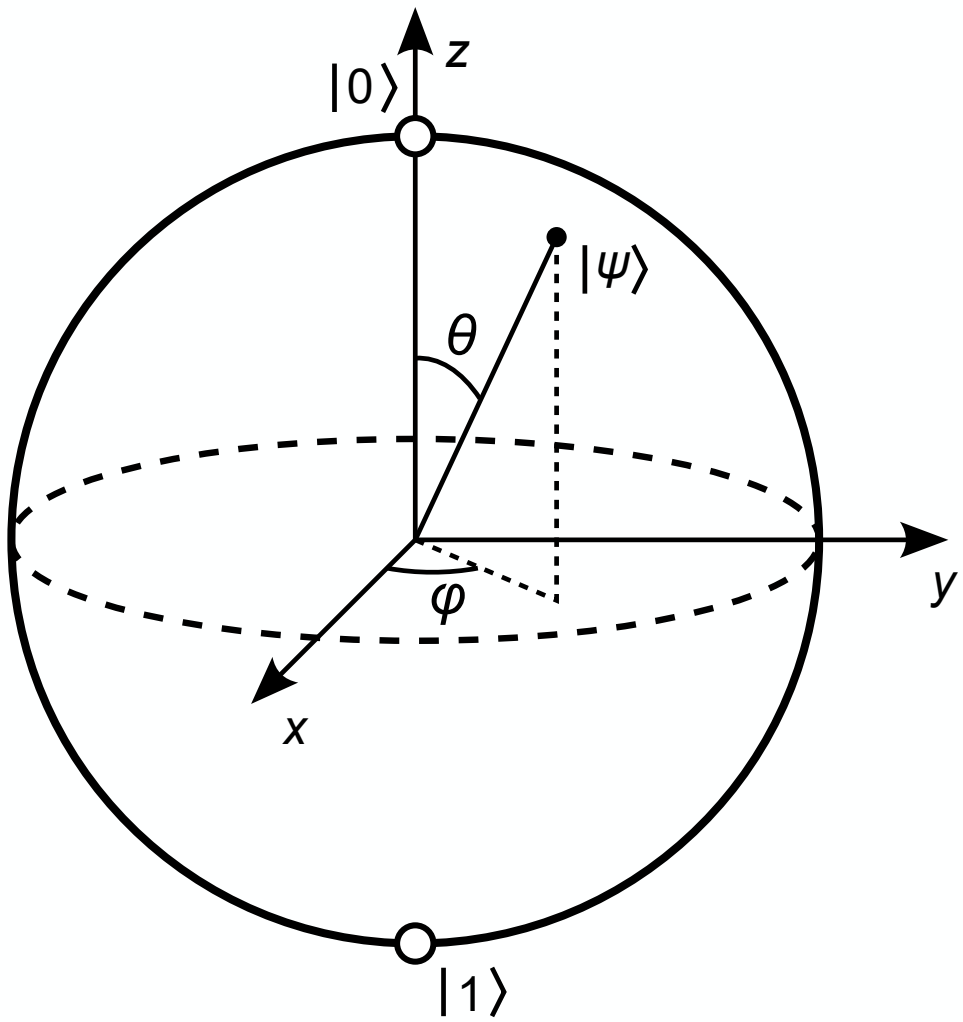

Figure 5: Bloch sphere representation[1]

PURDUE UNIVERSITY

What is a Magneto-Optical Trap?

- **Anti-Helmholtz** coils create a **quadrupolar** magnetic field
- **Detuned** beams apply a force to the atoms proportional to their velocity
- The combination of these two effects creates **optical molasses**
- These are a source of **entanglement** for the distributed quantum computing architecture

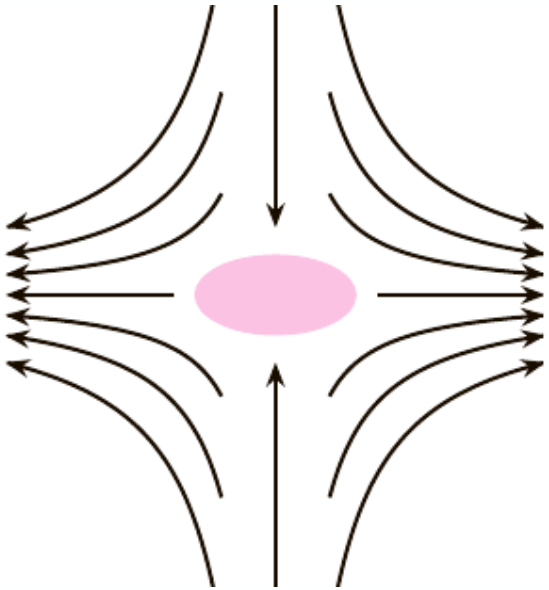

Figure 6: Quadrupolar magnetic field representation[3]

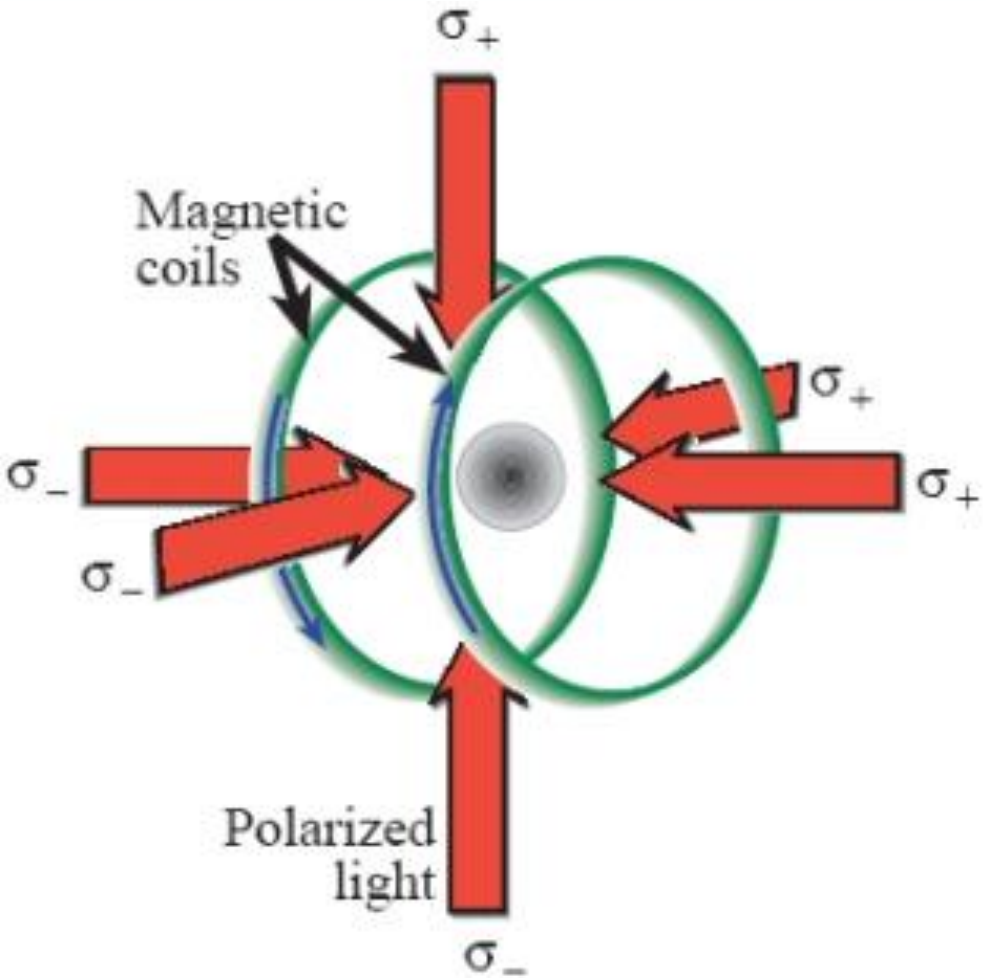

Figure 7: Standard 6 beam MOT setup[3]

PURDUE UNIVERSITY

Methods

PURDUE UNIVERSITY®

MOT Construction

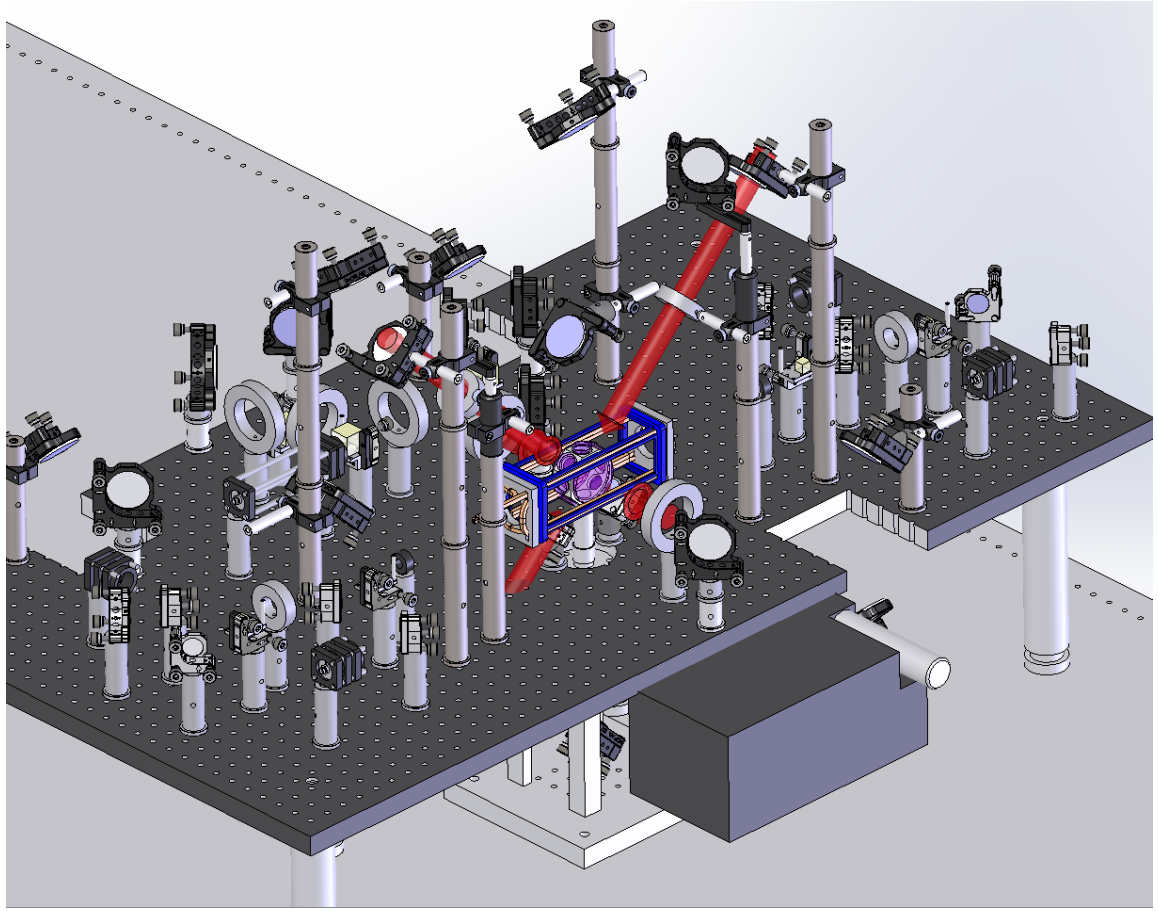

Figure 8: Rendering of 2D MOT setup

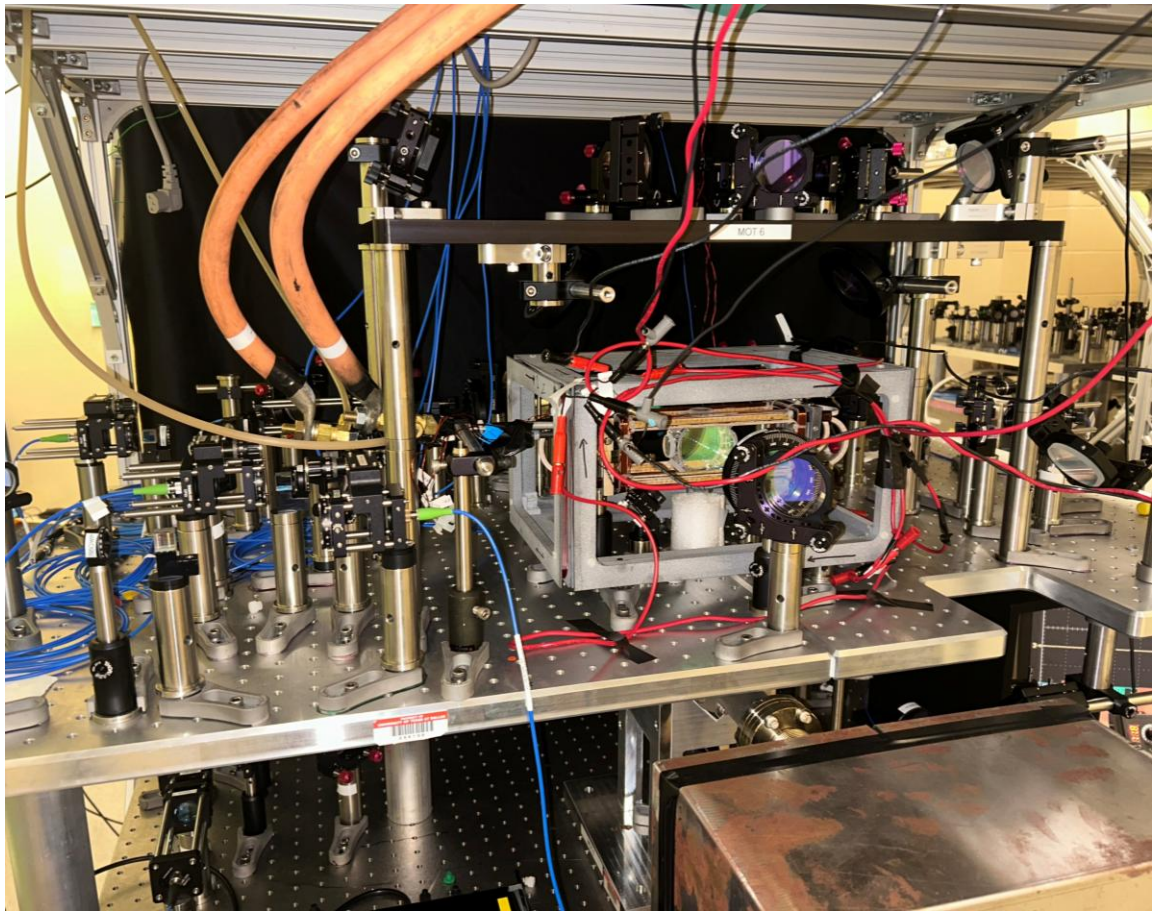

Figure 9: Implementation of 2D MOT setup

PURDUE UNIVERSITY

Measuring Optical Depth and Loading Time

- **Optical depth** measures light absorbed by MOT
- Higher optical depth yields better entanglement
- **Loading time** measures how fast the Rb atoms are slowed by the trapping scheme

$$\frac{dN}{dt} = R - \frac{N}{\tau}$$

Equation 1: Loading Time

$$OD = n\sigma L$$

Equation 2: Optical Depth

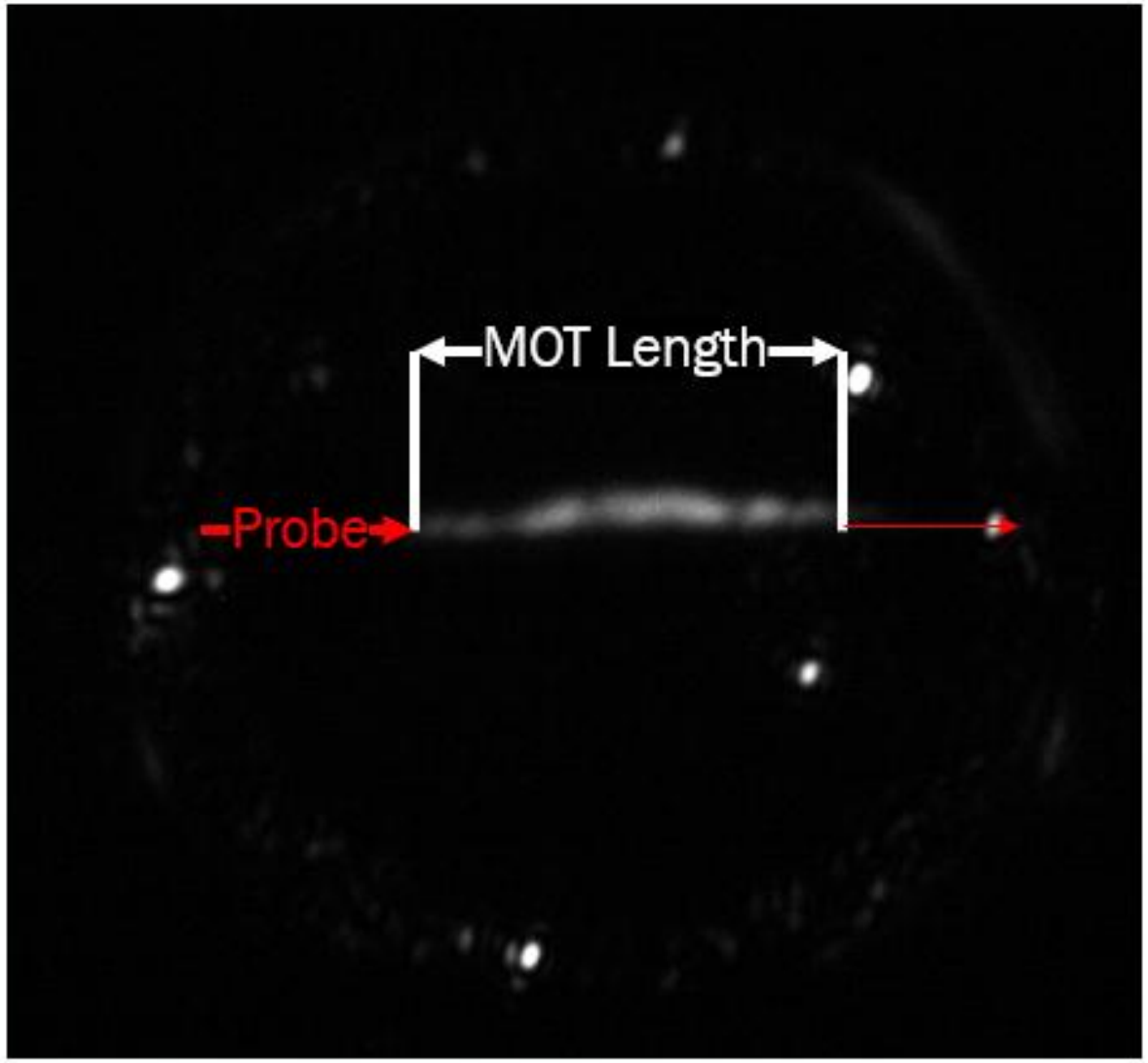

Figure 10: MOT Image with probe representation

PURDUE UNIVERSITY

Results

PURDUE UNIVERSITY®

MOT Parameters

- Trapping Power: 91.6mW
- Repump Power: 14.8mW
- Optical Depth: 40.2
- Loading Time: 3.2 seconds
- MOT Length: 0.8 cm

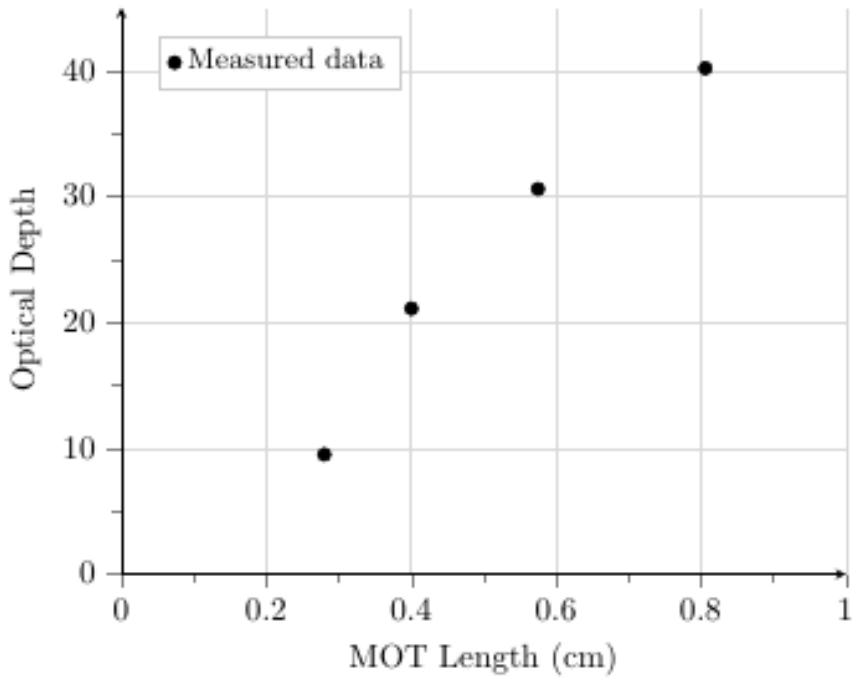

Figure 11: Changing MOT Length affects Optical depth

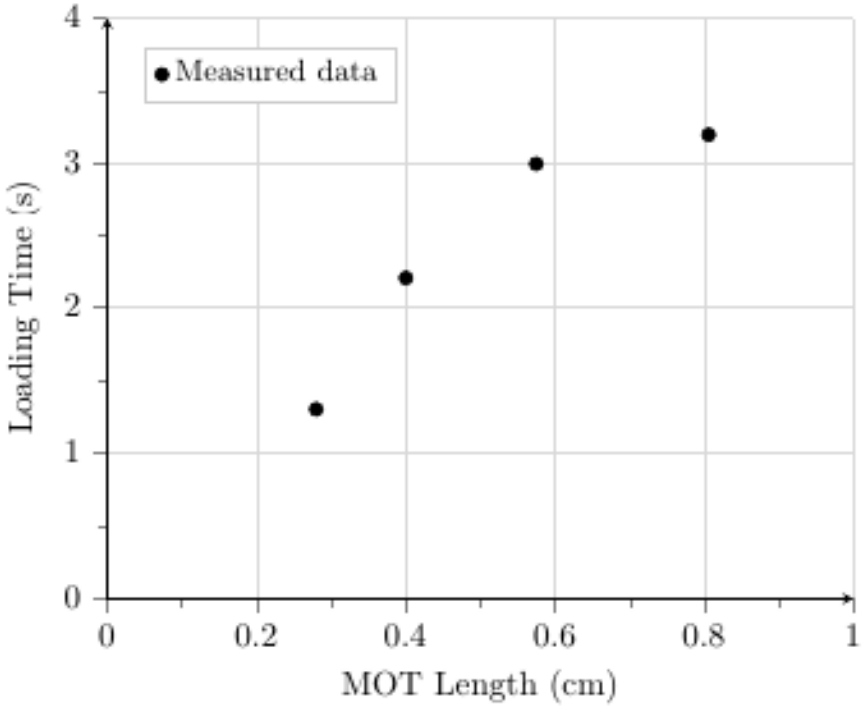

Figure 12: Changing MOT Length also affects Loading Time

PURDUE UNIVERSITY

Experiment Photos

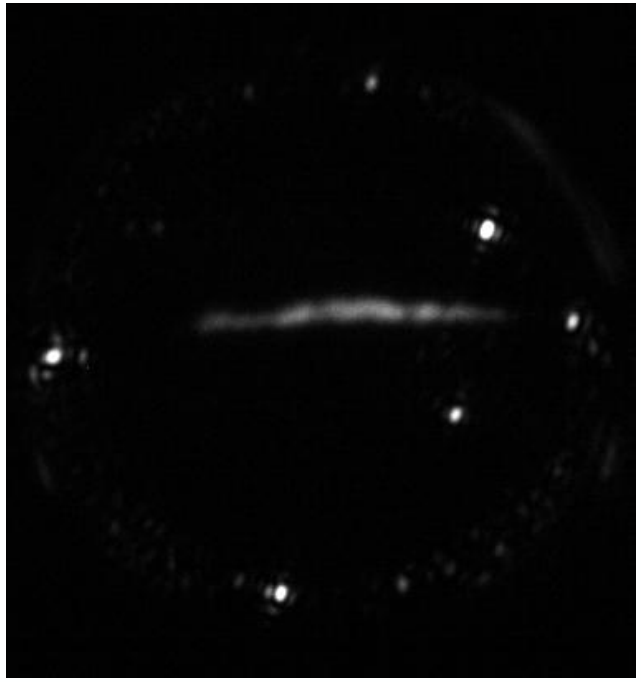

Figure 13: MOT Image
3A, Full Length

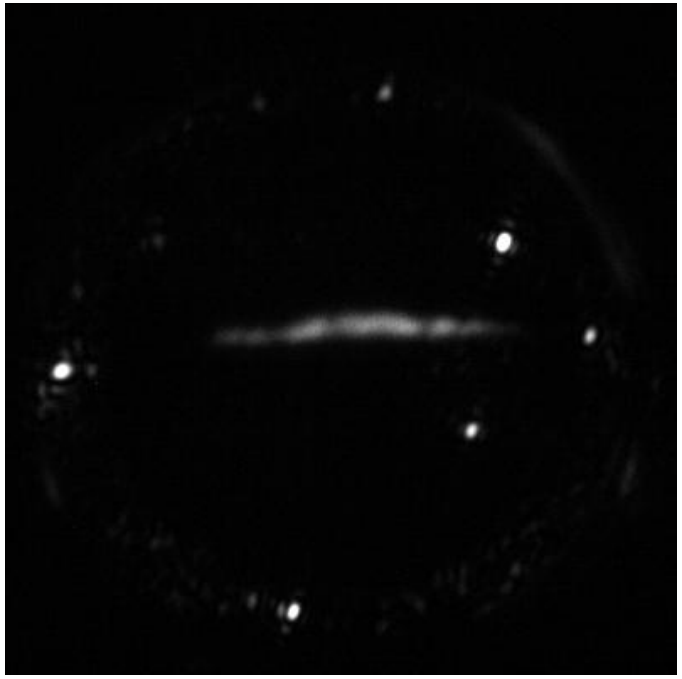

Figure 14: MOT Image
0A, Full Length

Figure 15: MOT Image
0A, Partial Length

PURDUE UNIVERSITY

Magnetic Confinement

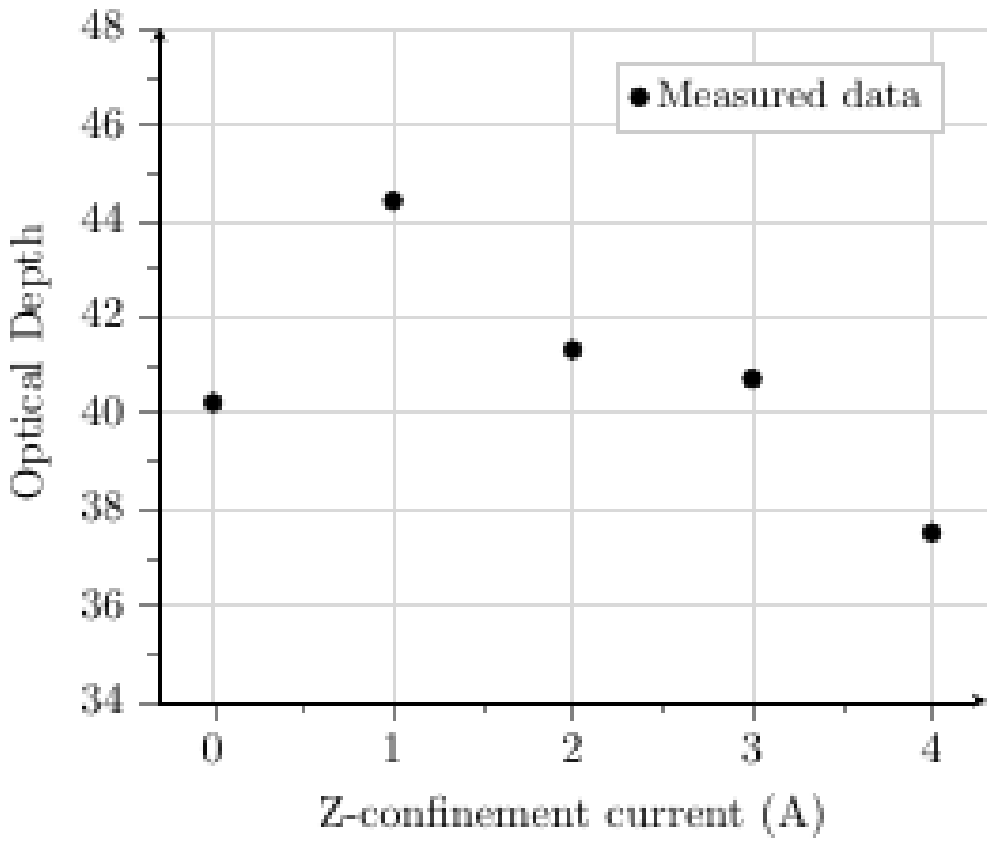

Figure 16: Optical Depth increases at some optimal confinement field

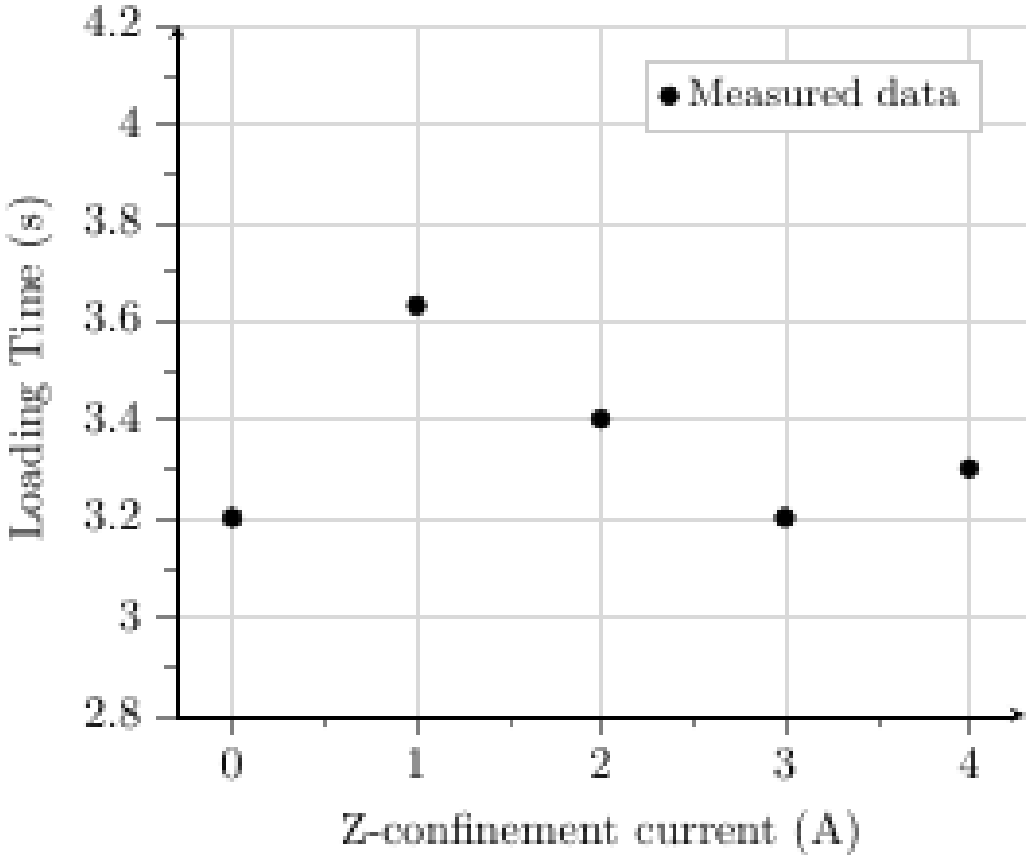

Figure 17: Loading time peaks at similar confinement field

PURDUE UNIVERSITY

Next Steps

PURDUE UNIVERSITY®

Polarization Entanglement

- 2 Photons input to the 2D MOT
- Entangled photons are then output
- Entanglement converted to **polarization**
- Higher optical depth creates better entanglement

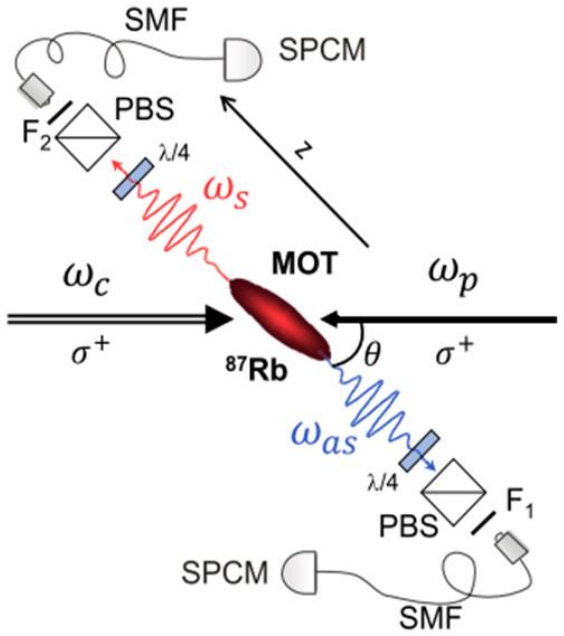

Figure 19: SFWM biphoton generation scheme[9]

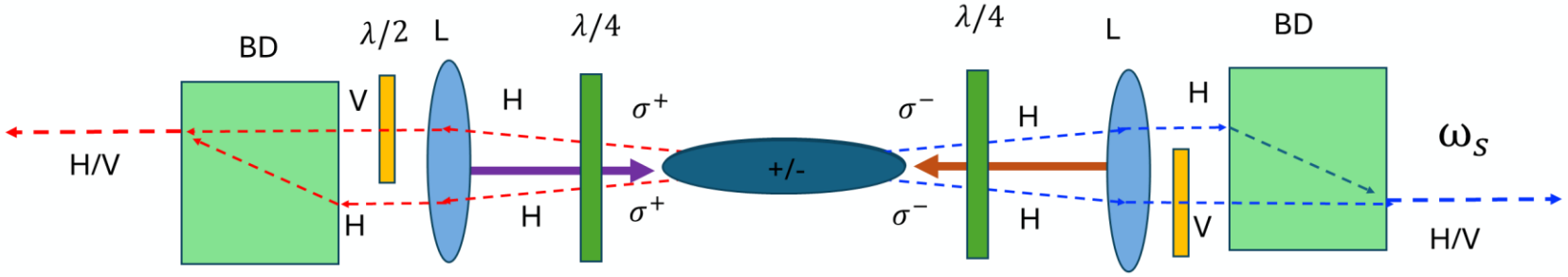

Figure 18: Optical setup for biphoton generation

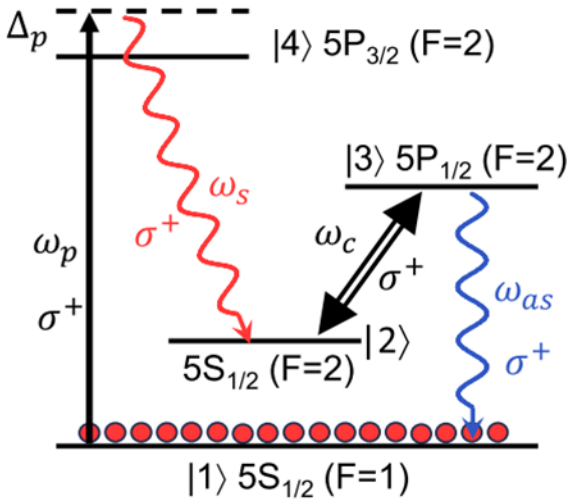

Figure 20: Optical 2-level system[9]

PURDUE UNIVERSITY

Works Cited

1: Smite-Meister, CC BY-SA 3.0, via Wikimedia Commons

2: Du, S., Ding, Y., & Qiao, C. (2025). S-QGPU: Shared quantum gate processing unit for distributed quantum computing. *AVS Quantum Science*, *7*, 013803. https://doi.org/10.1116/5.0241972

3: Du, S. (2005). *Atom-chip Bose-Einstein condensation in a portable vacuum cell* [Doctoral dissertation, University of Colorado Boulder]. JILA.

4: Raab, E. L., Prentiss, M., Cable, A., Chu, S., & Pritchard, D. E. (1987). Trapping of neutral sodium atoms with radiation pressure. *Physical Review Letters, 59*, 2631.

5: Chu, S. "The Manipulation of Neutral Particles." *Reviews of Modern Physics* 70, no. 3 (1998): 685–706.

6: Gustin, C., Dusanowski, Ł., Höfling, S., & Hughes, S. (2022). *Using the Autler-Townes and ac Stark effects to optically tune the frequency of indistinguishable single photons from an on-demand source*. *Physical Review Research, 4*, 023045.

7: Zhou MG, Cao XY, Lu YS, Wang Y, Bao Y, Jia ZY, Fu Y, Yin HL, Chen ZB. Experimental Quantum Advantage with Quantum Coupon Collector. Research (Wash D C). 2022 Apr 30;2022:9798679. doi: 10.34133/2022/9798679. PMID: 35586151; PMCID: PMC9082318.

8: Steck, D. A. (2003). *Rubidium 87 D Line Data* (Revision 1.6). Los Alamos National Laboratory.

9: Lai, X., Li, C., Zanders, A., Mei, Y., & Du, S. (2024). *Symmetry protected two-photon coherence time* (arXiv:2406.13028). arXiv

PURDUE UNIVERSITY

Questions?

PURDUE UNIVERSITY®

Optical Scheme

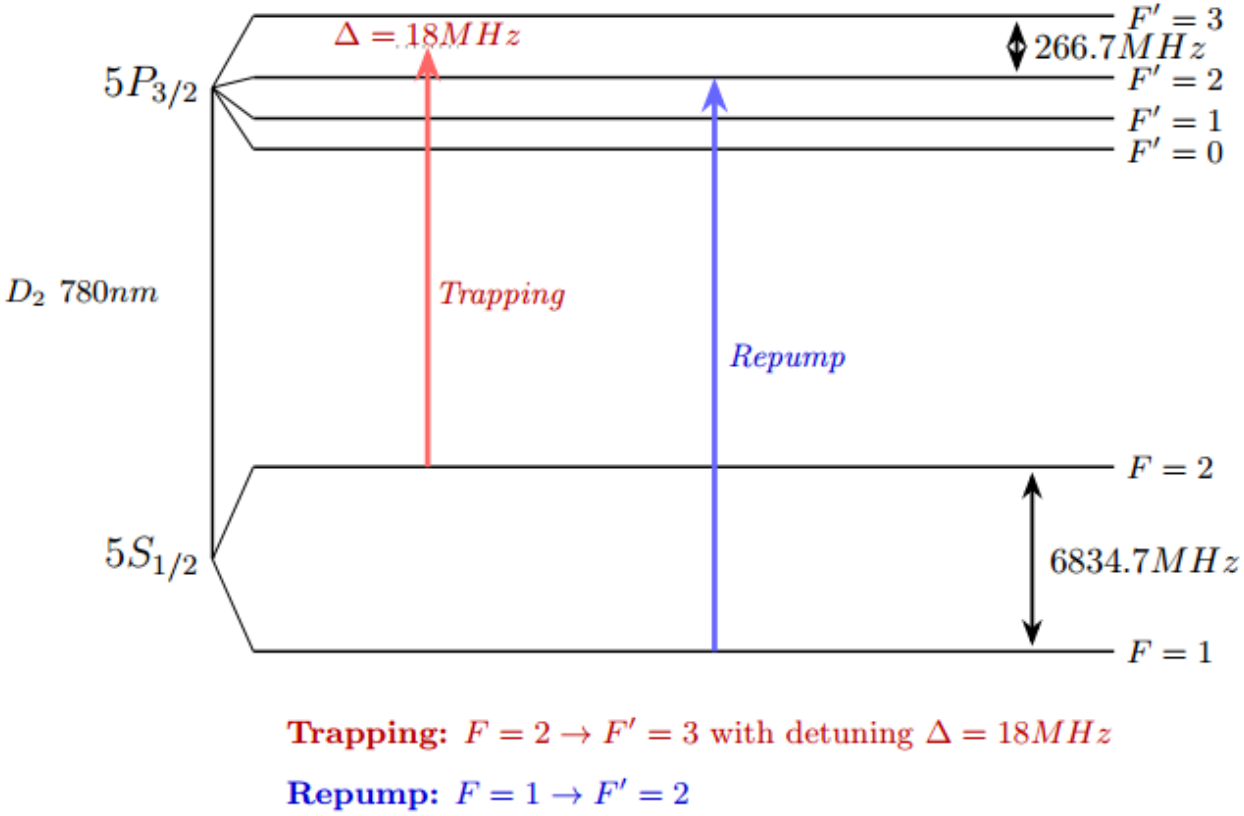

Figure A: Energy levels for 780nm laser[8]

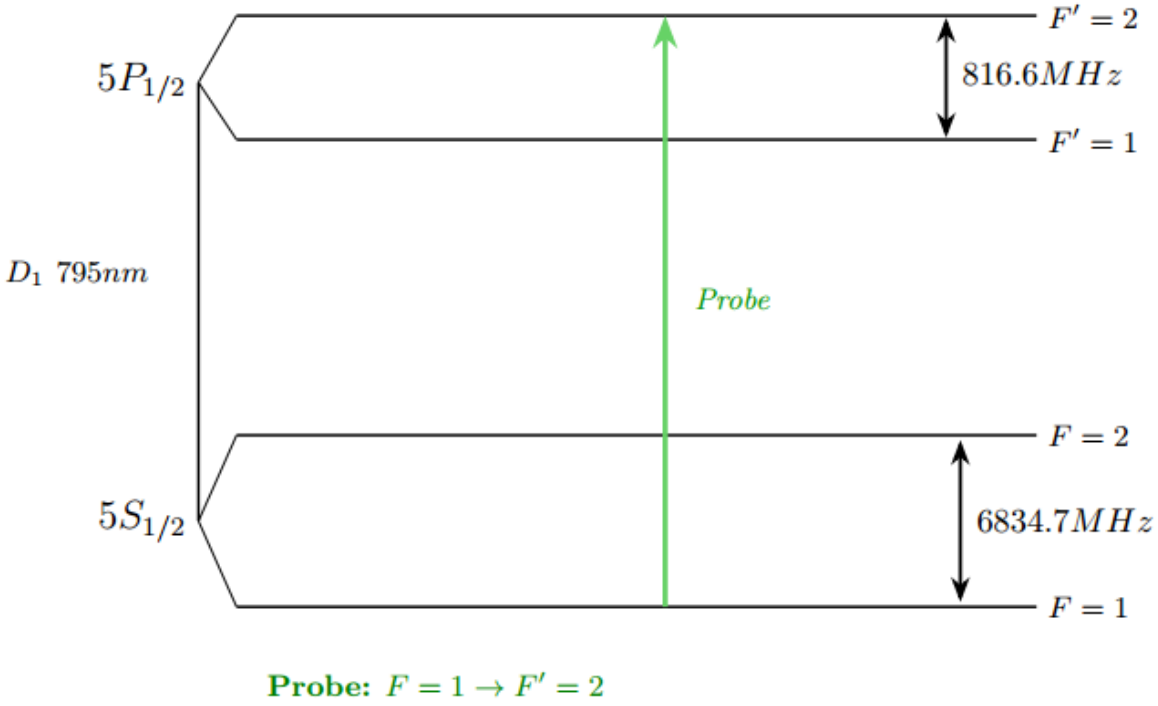

Figure B: Energy levels for 795nm laser[8]

PURDUE UNIVERSITY